

Electromyography-Based Hand Gesture Recognition System for Upper Limb Amputees

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Abstract—This article presents a low-power embedded platform that recognizes arm gestures by decoding surface electromyography (EMG) signals of amputees. The system consists of ADS1298 as analog front end (AFE), which can acquire 8-channel EMG signals simultaneously. For the validation of this system in real-time environment, four subjects have been recruited, including one transradial amputee. Six activities have been performed by each subject for the EMG pattern recognition experiment. Two cases have been created for testing data analysis of real-time classification in a DSP processor. Case one (feature extraction in DSP processor and testing off-line) shows the highest classification accuracy of 97.60% and the mean classification accuracy of 95.40%. Case two (training and testing within DSP processor) shows the highest classification accuracy of 97.75% and the mean classification accuracy of 92%. This shows that this system is capable of recognizing user intention in the real-time environment, with above 91% accuracy.

Index Terms—Sensor systems, amputees, classification, electromyography (EMG), hand gesture, real-time.

I. INTRODUCTION

Electromyography (EMG) signals are emerging as one of the indispensable biological parameters, which have extensive variety of utilizations in human-machine interaction, prosthetic device development, and rehabilitation devices. EMG measures muscle potential and can provide useful investigatory or diagnostic information regarding neuromuscular activity and rich motor control information for prosthetic devices. EMG signals can be decoded in order to control smart prosthetic devices using pattern recognition (PR) methods. This can be achieved by applying EMG electrodes on the residual limb, and a classification technique is utilized in the embedded platform to determine the user intention.

Many researchers have contributed in the EMG-based PR and control schemes. The EMG-based PR algorithm can be summarized as artificial neural network (ANN)-based algorithms, classifier-based algorithms, probabilistic model-based algorithms, and fuzzy logic-based algorithms [1], [2]. Neural network and linear discrimination analysis (LDA)-based techniques are best suitable schemes for implementing portable EMG-based gesture recognition system [3].

Portable EMG data acquisition system has also been developed for upper limb prosthesis application [4], [5], [10]. Targeted muscle reinnervation (TMR) techniques are also used for the extraction of user intention through invasive methods. Most of the proposed system perform off line training and are unable to detect transient response of the user [5]. There exists need to develop a real-time and portable system that can recognize neural information (EMG) for prosthetic applications. In the proposed work, a real-time PR platform has been developed for the prosthesis application. This system has eight differentials input ADC of 24 bit precision with lead off detection capability. An ARM Cortex M4 processor has been used for the DSP processing

in real time. The main contributions of the paper are as follows: (1) the proposed system performs online training of the user and predicts transient intention of the subject, (2) a flexible and adjustable arm band is designed for the forearm EMG signal acquisition, (3) this system has been tested on four subjects including a transradial amputee, and (4) a real-time embedded platform is developed to detect user intention in 75 ms, with above 91% accuracy.

The organization of paper is as follows: Section II describes the architecture of the proposed system and EMG signal acquisition protocol. In Section III, classification methodology and result have been discussed. The conclusion is presented in Section IV.

II. METHODOLOGY

A. System Architecture

System consists of ADS1298 as analog front end (AFE) or ADC, a DSP processor, a power management circuit, and an advanced and flexible band with adjustable strap. This band makes use of Ag-AgCl patches for EMG signal acquisition from the skin. At the input of the each channel, a protection circuitry and large biasing resistance (200 M Ω) have been utilized to maintain the input impedance as high. The common mode voltage of the circuit has been sent back to regulate the human body ground after a phase shift. A 32-bit DSP processor with ARM Cortex-M4 core is used to process algorithms digitally. The specification of AFE and DSP is given in the Table 1. The ADC sampling rate has been fixed at 1000 SPS for each eight differential input, and the system is powered by a 3.7-V lithium-ion battery.

For the training phase of the machine learning algorithm, a digital display is also incorporated with this system. The device weighs 17.5 g with battery support up to 40 hours of continuous operation. The system overview and arm band are shown in Figs. 1 and 2, respectively. Further, the 12-dB/oct Butterworth filter slope was selected because it has faster settling time and less overshoot. In this filter, 10–500 Hz

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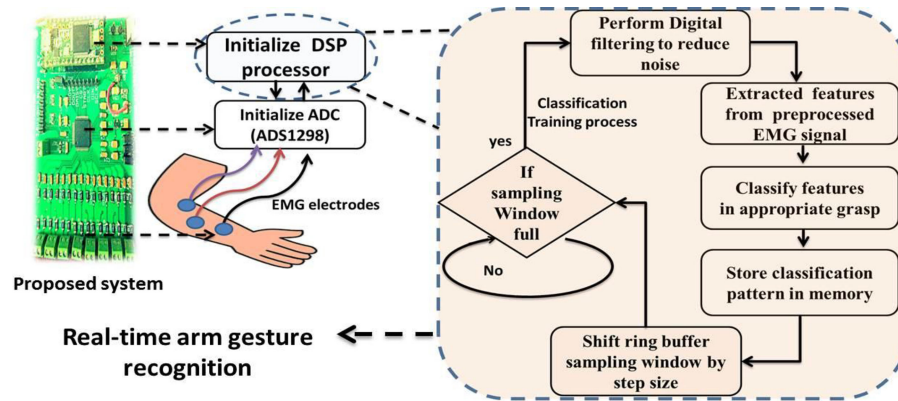


Fig. 1. System over view for the real time EMG activity classification of the transradial amputees.

Table 1. Specification of DSP Processor and Analog Front End

AFE	
Number of Channels	8 Differential inputs
Power Consumption	15 mW
Input noise	0.82 μ V
Acquisition frequency	1 KHz
DSP Processor	
Core Type	Arm Cortex M4
Voltage Range	1.71 to 3.6 V
Operating Frequency	150 MHz
Temperature Range	-40°C to 85°C

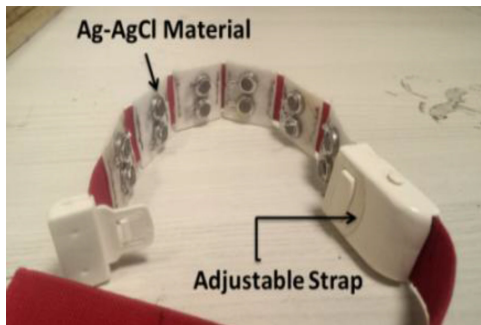


Fig. 2. Adjustable Ag-AgCl electrode band for the forearm.

band pass filter has been convolved with 50 Hz notch filter. Band pass filter coefficients have been computed in the MATLAB 2015a software. Circular buffers have been used for the buffering operations during signal processing in order to avoid overheads. In the proposed DSP processing, data from 8-channel differential input are collected and stored in the ring buffer after digitalization. From this buffer, data are utilized for feature extraction and subsequently the classifier uses these features for the training purpose of particular activity. The flow chart of LDA classifier implemented in the embedded system is shown in the Fig. 3.

B. Data Acquisition

For the Data acquisition, each amputee has been asked to sit on laboratory chair comfortably. Each subject has been requested to perform six grasping motions including one rest position, shown in Fig. 4.

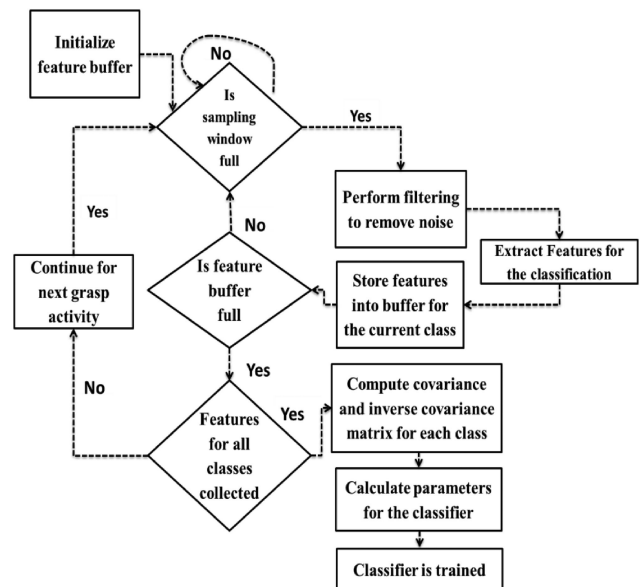


Fig. 3. Implementation of the LDA classifier algorithm for training in the DSP processor.

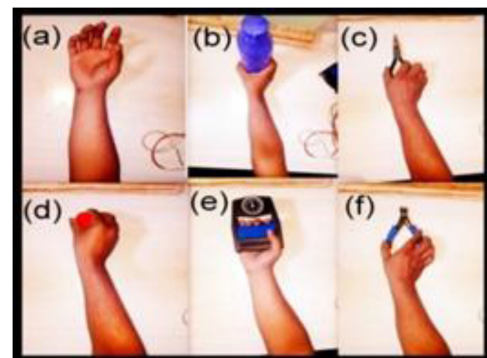


Fig. 4. Activities performed by the subjects including cylindrical and spherical grasp: (a) (rest), (b) (act 1), (c) (act 2), (d) (act 3), (e) (act 4), and (f) (act 5).

The arrangement of actions is not randomized in order to embolden the subjects to perform the movements as naturally as possible. The band has been placed around the forearm, which has six differential

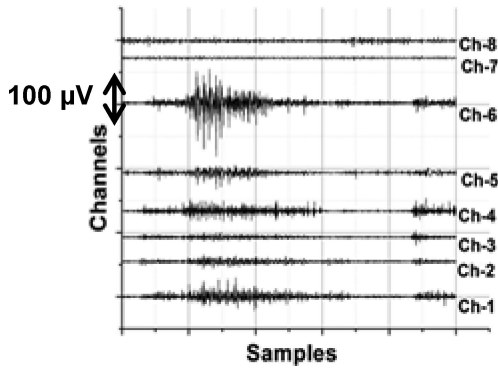


Fig. 5. EMG signals for the grasping activity of all eight channels.

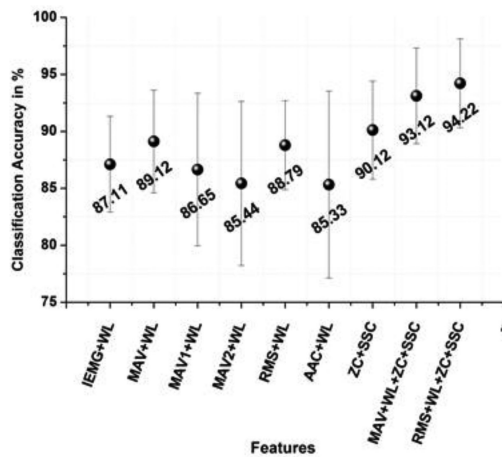


Fig. 6. Features examination for the system development.

Table 2. Classification Accuracy for the Case One

Sub	Subject Type	Classification accuracy in %
Sub1	Healthy	97.6%
Sub2	Healthy	94.5%
Sub3	Healthy	97.5%
Sub4	Amputee	92.0%

inputs, and two electrodes have been placed on the main activity spots of the biceps brachii and triceps brachii.

III. EXPERIMENT AND RESULTS

After calibration, the subject was requested to perform an online validation test, where each subject has been facilitated with computer display to see real-time activities. Filtered data from the system of all channels have been presented in the Fig. 5. For feature extraction, four features including root means square (RMS), waveform length (WL), zero crossing (ZC), and slope sign change (SSC) have been utilized. This feature set has been selected using our preliminary work performed off line and fused for the energy information (RMS and WL) and frequency information (SSC and ZC) presence in the EMG

Table 3. Classification Accuracy for the Case Two

Activities	Act.1 (Rest)	Act.2	Act.3	Act.4	Act.5	Act.6	Avg.
Subject							
Sub 1	100	100	86.6	93.3	90	100	94.98
Sub 2	100	100	96.3	96.6	93.3	100	97.75
Sub 3	100	93.3	86.6	90	90	93.3	92.20
Sub 4 (Amp.)	100	90	90	86.6	93.3	90	91.65

signal, shown in the Fig. 6. The window size of 200 ms has been taken using 75 ms increment [7], [8].

Further, two cases have been created for the performance analysis of the developed system. In case one, data have been acquired through the system for computation of features for training matrix preparation. These features have been saved in the memory for further analysis. Subsequently in case two, training and testing have been performed within DSP processor for hard real-time application.

A. Case One

For the first case, classification performance for each subject has been presented in the Table 2. The highest classification accuracy is achieved by the subject one and third (both are nonamputees). PR rate for amputee is 92%, which is appropriate for the real-time performance. The confusion matrices for all the subjects are presented in the Fig. 7. There are few misclassifications between act 4 and act 5.

B. Case Two

Each of subjects was instructed to perform each of the five grasping actions 30 times for the testing purpose. Transradial amputee who participated in this experiment is amputated more than 2.5 years and has not been using any kind of prosthetic aids. The activity completion time was fixed at 75 ms. The classification accuracy for each subject is presented in the Table 3. The mean classification accuracy achieved is 94.14% for all the subjects. Highest classification accuracy of 97.75% is achieved by subject 2. Amputee has achieved 91.65% classification accuracy, which shows that this system can work in the real-time environment for the transradial amputees.

Subsequently, comparison of embedded system parameter for real-time hand gesture recognition has been shown in Table 4. From the analysis, it is clear that the proposed system is capable to perform training and testing online, and the classification accuracy has been improved over the existing system described in the literature. The LDA classifier is used, which is much simpler than SVM classifier, because there no parametric change required. LDA depends only on feature vector size, whereas SVM depends on both feature vector size and support vectors. LDA has low computation complexity [7]. The advantage of using online training is that the current information related to EMG signal can be utilized, thereby enhancing the overall accuracy of the system. Another advantage of the implemented system is flexibility of the band used in the signal acquisition, which makes it feasible to be used for varying forearm dimensions and positions. This system also makes possible recognition of transient user intention. Power breakdown diagram is also presented in the Table 5, which shows that the average power consumption of the implemented

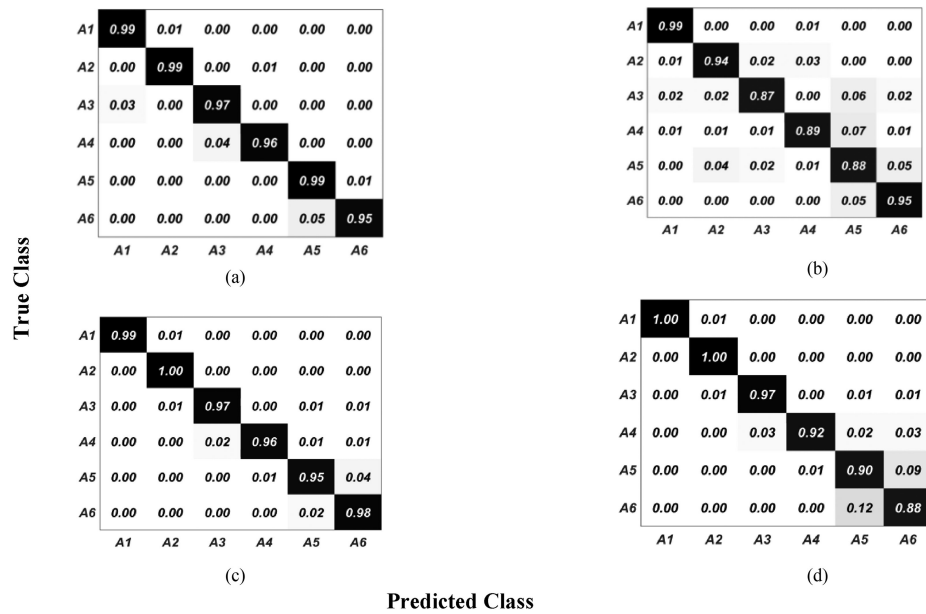


Fig. 7. Case one confusion matrices: (a) subject 1, (b) subject 2, (c) subject 3, and (d) subject 4.

Table 4. Comparison With Existing Systems

Parameter	Chen et al [1]	Benatti et al [2]	lee et al [9]	This work
System Implemented	Computer	Embedded	Computer	Embedded
Training	Off-line	Off-line	Off-line	On-line
Testing	On-line	On-line	On-line	On-line
Amputee considered	No	No	Yes	Yes
Classifier	SVM	SVM	ANN	LDA
Gesture State	Hold	Hold	Hold	Transient
Number of Gesture	9	7	6	6
Number of channel	4	8	8	8
Accuracy (mean)	90 %	90 %	93.20 %	94.14%
Power consumption	-	29.7 mW	-	26.6 mW

Table 5. Power Consumption for the Proposed System

S. No.	Parameters	Power consumption
1.	SPI read UART send	35.2 mW
2.	EMG signal preprocessing	20.3 mW
3.	Feature extraction	40.3 mW
4.	classification	26.6 mW

system is in the same range as that of the existing systems being compared.

IV. CONCLUSION

In this article, a real-time arm gesture recognition system is presented for the amputees. This system can be utilized for myoelectric prosthetic control in the real-time environment, with above 91% of accuracy. Four time domain features and LDA classifier are incorporated within the DSP processor. With this system, grasping motion can decode easily in 75 ms completion time after training model generation. This system has light weight and battery power support up to 40 Hrs. This system presents admirable results for low-power applications in rehabilitation and myoelectric prosthetics. Power optimization will further enhance the life time of the proposed system.

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